

AWS TOPIC WISE INTERVIEW QUESTIONS

CLOUD COMPUTING

◇ Basic Level

These test fundamental understanding.

1. What is cloud computing?

- *Answer:* Delivery of computing services (like servers, storage, databases, networking, software) over the internet ("the cloud") to offer faster innovation, flexible resources, and economies of scale.

2. What are the main types of cloud computing?

- *Answer:*
 - IaaS (Infrastructure as a Service)
 - PaaS (Platform as a Service)
 - SaaS (Software as a Service)

3. What are the deployment models in cloud computing?

- *Answer:*

- Public Cloud
- Private Cloud
- Hybrid Cloud
- Community Cloud

4. Name some popular cloud service providers.

- *Answer:* Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP), IBM Cloud, Oracle Cloud.

5. What are some key benefits of cloud computing?

- *Answer:* Scalability, cost-efficiency, flexibility, disaster recovery, global reach, security, and faster deployment.

♦ Intermediate Level

These go deeper into concepts and real-world usage.

6. What is virtualization in cloud computing?

- *Answer:* It's the creation of virtual versions of physical components (servers, storage, networks) to increase efficiency and scalability.

7. What is autoscaling?

- *Answer:* Automatic adjustment of compute resources based on demand to optimize cost and performance.

8. **Explain the concept of multi-tenancy in the cloud.**

- *Answer:* A single instance of software serves multiple customers (tenants), each isolated from the other.

9. **What is the difference between vertical and horizontal scaling?**

- *Answer:*
 - *Vertical Scaling:* Increasing power (CPU/RAM) of a single server.
 - *Horizontal Scaling:* Adding more servers to distribute the load.

10. **What is a load balancer and why is it used?**

- *Answer:* Distributes incoming traffic across multiple resources to ensure availability and performance.

11. **What are containers in cloud computing?**

- *Answer:* Lightweight, portable units (like Docker) for deploying applications consistently across environments.

12. **What's the difference between containers and virtual machines (VMs)?**

- *Answer:* Containers share the host OS and are more lightweight; VMs emulate full hardware and include guest OS.

△ Advanced Level

These are more technical and architecture-focused.

13. What are microservices and how do they relate to the cloud?

- *Answer:* Microservices are small, independent services that work together in a cloud environment to form a complete application.

14. How does cloud computing enhance disaster recovery?

- *Answer:* Cloud providers offer replication, backup, and geographic redundancy for faster recovery and lower costs.

15. Explain the CAP Theorem and its relevance to cloud databases.

- *Answer:* A distributed system can only guarantee two of the following three: Consistency, Availability, Partition Tolerance.

16. How does a CDN (Content Delivery Network) work in the cloud?

- *Answer:* Delivers content via a network of edge servers located worldwide to reduce latency and improve load times.

17. What are some cloud security best practices?

- *Answer:* Identity and Access Management (IAM), encryption, multi-factor authentication (MFA), security groups/firewalls, logging/monitoring.

18. **What is serverless computing?**

- *Answer:* A cloud execution model where the provider dynamically manages server allocation. Example: AWS Lambda, Azure Functions.

19. **How do you ensure high availability in cloud architecture?**

- *Answer:* Use load balancing, failover mechanisms, multi-AZ/region deployment, redundant systems, and autoscaling.

20. **What is Infrastructure as Code (IaC)?**

- *Answer:* Managing and provisioning infrastructure through code using tools like Terraform, AWS CloudFormation, or Azure Resource Manager.

AWS GLOBAL INFRASTRUCTURE

♦ Basic Level

1. **What is AWS Global Infrastructure?**

- *Answer:* AWS Global Infrastructure refers to the globally distributed data centers, including Regions, Availability Zones (AZs), Edge Locations, and Local Zones.

2. **What is a Region in AWS?**

- *Answer:* A Region is a geographical area with at least two Availability Zones. Examples include **us-east-1** (N. Virginia) and **eu-west-1** (Ireland).

3. What is an Availability Zone?

- *Answer:* An AZ is one or more discrete data centers with redundant power, networking, and connectivity within a Region.

4. What is an Edge Location?

- *Answer:* These are endpoints for AWS services (like CloudFront) to deliver low-latency content to users.

5. What is the difference between an AZ and a Region?

- *Answer:* A Region is a broad geographic area, while AZs are isolated locations within Regions to enable high availability.

◇ Intermediate Level

6. How does AWS ensure high availability and fault tolerance using AZs?

- *Answer:* By deploying applications across multiple AZs, if one fails, others can take over with minimal disruption.

7. What are Local Zones and why are they used?

- *Answer:* Local Zones place compute, storage, and database services closer to end users in specific locations to reduce latency.

8. How many Regions and AZs does AWS currently have?

- *Answer:* You'll need to check the latest numbers on the [AWS Infrastructure page](#) as they change frequently.

9. How does AWS Direct Connect fit into the global infrastructure?

- *Answer:* It provides a dedicated network connection between your premises and AWS, improving performance and reducing latency.

10. Can resources be shared across Regions?

- *Answer:* Generally no, but some services like Amazon S3, IAM, and CloudFront have global scope or replication features across regions.

△ Advanced/Architectural Level

11. How would you design a multi-region application on AWS?

- *Answer:* Use Route 53 latency-based routing, S3 cross-region replication, multi-region DynamoDB Global Tables, and decouple using services like SNS/SQS.

12. What are the challenges in implementing cross-region replication?

- *Answer:* Latency, data consistency, increased cost, complexity in failover mechanisms, and compliance with data residency laws.

13. How does AWS support compliance with local data laws globally?

- *Answer:* By allowing users to choose Regions for resource deployment and not moving data out of the selected Region without user action.

14. What role do AWS Wavelength and Outposts play in the global infrastructure?

- *Answer:* Wavelength brings services closer to 5G devices for ultra-low latency; Outposts extend AWS infrastructure on-premise.

15. Explain a disaster recovery strategy using AWS Regions and AZs.

- *Answer:* Use strategies like Pilot Light or Warm Standby with cross-region replication of data and services, leveraging Route 53 failover routing.

AWS CONSOLE

♦ Basic Level

1. What is the AWS Management Console?

- *Answer:* It's a web-based interface that allows users to manage AWS services and resources using a graphical interface.

2. How do you navigate to a specific AWS service in the Console?

- *Answer:* Use the search bar at the top of the console or browse services categorized by function (e.g., Compute, Storage, Networking).

3. What are AWS regions and availability zones, and how do you select them in the Console?

- *Answer:* You can select a region from the top-right dropdown menu. Resources are created in the selected region.

4. Can you access the AWS CLI or SDK from the Console?

- *Answer:* No. The AWS Console is for GUI access; the CLI and SDKs are used via terminal/code.

5. What is the difference between the AWS Console and AWS CLI?

- *Answer:* The Console is graphical; CLI is command-line. CLI allows automation and scripting, whereas the Console is used for manual tasks and visualization.

◇ Intermediate Level

6. How can you view billing and usage details in the Console?

- *Answer:* Navigate to Billing and Cost Management Dashboard, accessible from the Services menu or your account dropdown.

7. What is IAM, and how do you manage users and permissions via the Console?

- *Answer:* IAM (Identity and Access Management) allows you to manage users, roles, and permissions via a dedicated IAM Console section.

8. What steps do you take to launch an EC2 instance from the Console?

- *Answer:* Go to EC2 > Launch Instance > Choose AMI > Select instance type > Configure > Add storage > Add tags > Configure security group > Launch.

9. How can you monitor resource usage via the Console?

- *Answer:* Use CloudWatch Dashboards, EC2 metrics, S3 usage statistics, or Service Quotas.

10. How do you enable or disable Multi-Factor Authentication (MFA) via the AWS Console?

- *Answer:* Go to IAM > Users > Select a user > Security Credentials > Manage MFA.

△ Advanced Level

11. How do you configure a VPC using the AWS Console?

- *Answer:* Go to VPC > Launch VPC Wizard or use step-by-step creation of subnets, route tables, gateways, etc.

12. How would you configure S3 bucket policies via the Console?

- *Answer:* Open the bucket > Permissions tab > Bucket policy editor to define or edit JSON-based policies.

13. How can you automate deployments from the AWS Console?

- *Answer:* Use AWS CloudFormation, Elastic Beanstalk, or CodePipeline, all available via the Console.

14. How do you manage cross-region replication in S3 from the Console?

- *Answer:* Go to S3 > Select bucket > Management > Replication > Add rule > Define destination bucket and permissions.

15. What steps would you take to set up a CloudFront distribution via the Console?

- *Answer:* Go to CloudFront > Create Distribution > Choose source (e.g., S3) > Configure settings (cache, origin, SSL) > Create.

⚡ Bonus: Scenario-Based Questions

16. You accidentally deleted a resource using the AWS Console. How do you recover it?

- *Answer:* Recovery options depend on the service. For example, use S3 versioning or RDS snapshots if previously enabled.

17. How would you troubleshoot a failed Lambda function using the Console?

- *Answer:* Navigate to Lambda > Select function > Monitor tab > View CloudWatch logs.

18. Your EC2 instance is not accessible. How would you debug this using the Console?

- *Answer:* Check security groups, instance status, system logs, network ACLs, and route tables.

19. **How can you set budget alerts via the AWS Console?**

- *Answer:* Go to Billing > Budgets > Create budget > Set threshold and notification.

20. **How do you create a custom dashboard to monitor multiple AWS services?**

- *Answer:* Use CloudWatch > Dashboards > Create Dashboard > Add widgets for metrics like EC2, S3, Lambda, etc.

EC2 Instance Interview Questions

1. **What is an EC2 instance in AWS?**

- *Answer:* Amazon EC2 (Elastic Compute Cloud) is a web service that provides resizable compute capacity in the cloud, allowing users to run virtual servers.

2. **What are the steps to launch an EC2 instance?**

- *Answer:*

- Choose AMI

- Choose instance type
- Configure instance details
- Add storage
- Add tags
- Configure security group
- Review and launch with a key pair

3. What types of storage options are available for EC2?

- *Answer:* Amazon EBS (block storage), Instance Store (ephemeral), and EFS (shared file storage).

4. Can we stop and start a spot instance?

- *Answer:* No, spot instances are terminated when stopped or interrupted.

5. How can you connect to an EC2 instance?

- *Answer:* Via SSH (Linux) or RDP (Windows); use PuTTY for Windows-based SSH login if .ppk format is required.

6. What is PuTTY and why is it used with AWS EC2?

- *Answer:* PuTTY is an SSH client for Windows used to connect to Linux EC2 instances. It uses .ppk key files.

7. How do you convert a .pem file to .ppk?

- *Answer:* Use PuTTYgen tool: Load .pem → Save private key as .ppk.

8. What might cause a PuTTY connection to EC2 to fail?

- *Answer:*
 - Incorrect key file
 - Wrong user (ec2-user, ubuntu, etc.)
 - Port 22 not open in Security Group
 - Public IP address changed (elastic IP not assigned)
 - Instance state not running

9. Which user names are used for different EC2 AMIs?

- *Answer:*

- Amazon Linux/RedHat → **ec2-user**
 - Ubuntu → **ubuntu**
 - Debian → **admin** or **root**
 - CentOS → **centos**
-

◇ Launching Multiple EC2 Instances Interview Questions

10. How do you launch multiple EC2 instances at once?

- *Answer:* During the instance launch process in the AWS Console or CLI, set the “Number of instances” field.

11. Are all instances in a multiple-instance launch identical?

- *Answer:* Yes, they use the same AMI, type, key pair, security group, and other settings unless configured differently via automation or scripting.

12. Can you assign different key pairs or security groups to each instance in a bulk launch via console?

- *Answer:* No, you’ll need to configure them individually after launch or use automation (e.g., CloudFormation or scripts).
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◇ Password vs Key-Based Authentication Interview Questions

13. What is key-based authentication?

- *Answer:* Authentication using a public-private key pair. The private key stays with the user; the public key is stored on the server (e.g., EC2).

14. What are the advantages of key-based over password-based authentication?

- *Answer:*
 - More secure (not susceptible to brute-force attacks)
 - Automatable
 - No password sharing
 - Easier to manage access via key rotation

15. Can EC2 use password authentication by default?

- *Answer:* No. AWS disables password authentication for Linux EC2 instances by default for security reasons. You can enable it by manually setting a password and editing the SSH config.

16. How would you enable password authentication on a Linux EC2 instance?

- *Answer:*
 - SSH into instance

- Edit `/etc/ssh/sshd_config` and set `PasswordAuthentication` yes
- Restart SSH service
- Set a password for the user using `passwd` command

17. Is it safe to use password-based authentication on public cloud servers?

- *Answer:* Not recommended. Key-based authentication is preferred due to better security and automation support.

⚡ Scenario-Based Questions

18. You lost the `.pem` key for an EC2 instance. How do you access the instance?

- *Answer:*
 - Stop the instance
 - Detach the root volume
 - Attach it to a temporary instance
 - Mount and modify the `authorized_keys`
 - Reattach to original instance

19. How would you automate launching and configuring multiple EC2 instances?

- *Answer:* Use AWS CLI, CloudFormation, or Terraform. You can specify instance count, tags, security groups, and startup scripts (User Data).

20. **How do you rotate EC2 SSH key pairs securely?**

- *Answer:*
 - Add new public key to `~/.ssh/authorized_keys`
 - Test login
 - Remove old key
 - Optionally update associated key pair metadata in AWS

Basic Level Networking Questions

1. **What is a network?**

- *Answer:* A network is a group of two or more connected devices that share resources and data.

2. **What are the different types of networks?**

- *Answer:* LAN (Local Area Network), WAN (Wide Area Network), MAN (Metropolitan Area Network), PAN (Personal Area Network).

3. **What is an IP address?**

- *Answer:* An Internet Protocol (IP) address is a unique identifier assigned to a device on a network.

4. What is the difference between IPv4 and IPv6?

- *Answer:*
 - IPv4 is 32-bit (e.g., 192.168.0.1), IPv6 is 128-bit (e.g., 2001:db8::1).
 - IPv6 supports vastly more addresses and includes security and routing improvements.

5. What is a MAC address?

- *Answer:* A Media Access Control address is a hardware identifier assigned to a network interface card (NIC).

6. What is the difference between a hub, switch, and router?

- *Answer:*
 - *Hub:* Broadcasts data to all ports.
 - *Switch:* Sends data to the intended recipient using MAC address.
 - *Router:* Connects different networks and routes packets using IP addresses.

7. What is DNS and how does it work?

- *Answer:* Domain Name System resolves domain names (e.g., google.com) to IP addresses.
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◇ Intermediate Level Networking Questions

8. What is the OSI model? Name its layers.

- *Answer:* OSI (Open Systems Interconnection) model is a framework for understanding network interactions.
 - Layers: Physical, Data Link, Network, Transport, Session, Presentation, Application

9. What is the difference between TCP and UDP?

- *Answer:*
 - TCP: Connection-oriented, reliable, slower (e.g., HTTP, FTP)
 - UDP: Connectionless, faster, less reliable (e.g., DNS, video streaming)

10. What is subnetting?

- *Answer:* The process of dividing a network into smaller sub-networks to improve performance and security.

11. What is the default gateway?

- *Answer:* It's the device (usually a router) that connects a local network to external networks.

12. What are private and public IP addresses?

- *Answer:*
- *Private IPs* are used within a network (e.g., 192.168.x.x).
- *Public IPs* are assigned by ISPs and are accessible over the internet.

13. What is NAT (Network Address Translation)?

- *Answer:* NAT translates private IP addresses to a public IP for internet access.

14. What is a firewall?

- *Answer:* A network security device or software that filters incoming and outgoing traffic based on rules.

△ Advanced/Scenario-Based Questions

15. What is DHCP and how does it work?

- *Answer:* Dynamic Host Configuration Protocol automatically assigns IP addresses to devices in a network.

16. What is ARP?

- *Answer:* Address Resolution Protocol maps IP addresses to MAC addresses in a local network.

17. What is the difference between static and dynamic IP addressing?

- *Answer:*
- *Static:* Manually assigned and doesn't change
- *Dynamic:* Automatically assigned by DHCP and can change

18. What is a VLAN?

- *Answer:* A Virtual LAN allows logical segmentation of a network at the switch level, increasing security and traffic management.

19. How does a 3-way TCP handshake work?

- *Answer:*
- SYN → SYN-ACK → ACK
- Establishes a reliable connection between client and server.

20. How do you troubleshoot a network that is down?

- *Answer:*

- Check physical connections
- Use **ping**, **tracert**, or **ipconfig/ifconfig**
- Check IP settings
- Check router/modem logs
- Validate DNS and gateway settings

⚡ Bonus: Quick Definitions

Term	Description
Ping	Tests connectivity between hosts
Traceroute	Shows path of packet to destination
Loopback Address	127.0.0.1 – used to test local TCP/IP stack
Port Number	Identifies specific processes/services (e.g., 80 for HTTP)
Bandwidth vs Latency	Bandwidth = volume, Latency = delay

☑ Basic IP Addressing Interview Questions

1. **What is an IP address?**

- Answer: An IP address (Internet Protocol address) is a unique identifier assigned to each device on a network to allow communication between them. It can be IPv4 (e.g., 192.168.1.1) or IPv6 (e.g., 2001:db8::1).

2. What are the two versions of IP addresses?

- Answer:
 - IPv4: 32-bit address, uses 4 octets (e.g., 192.0.2.1)
 - IPv6: 128-bit address, uses 8 groups of hexadecimal numbers (e.g., 2001:0db8:85a3::8a2e:0370:7334)

3. What are the different classes of IPv4 addresses?

- Answer:

Class	Range	Default Subnet Mask	Use
A	1.0.0.0 – 126.255.255.255	255.0.0.0	Large networks
B	128.0.0.0 – 191.255.255.255	255.255.0.0	Medium networks
C	192.0.0.0 – 223.255.255.255	255.255.255.0	Small networks
D	224.0.0.0 – 239.255.255.255	N/A	Multicast

E	240.0.0.0 – 255.255.255.255	N/A	Reserved
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Note: 127.x.x.x is reserved for loopback addresses.

☒ Public vs Private IP Interview Questions

4. What is a public IP address?

- Answer: An IP address that is routable on the internet. Assigned by an ISP and globally unique.

5. What is a private IP address?

- Answer: An IP used within private networks, not routable on the public internet.

Class	Private IP Range
A	10.0.0.0 – 10.255.255.255
B	172.16.0.0 – 172.31.255.255
C	192.168.0.0 – 192.168.255.255

6. Can a private IP access the internet directly?

- Answer: No. It must go through NAT (Network Address Translation) to use a public IP.

7. How do you check if an IP is public or private?

- Answer: By comparing it to the reserved private ranges above or using tools like `ipconfig`, `ifconfig`, or public IP checkers online.
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☒ Static vs Dynamic IP Interview Questions

8. What is a static IP address?

- Answer: An IP address manually assigned to a device that does not change.

9. What is a dynamic IP address?

- Answer: An IP assigned by a DHCP server that can change over time.

10. What are the advantages of static IPs?

- Answer:
 - Consistent access (ideal for servers, printers, remote access)
 - Easier DNS mapping
 - More predictable for security/firewall rules

11. What are the disadvantages of static IPs?

- Answer:
 - Manual configuration overhead

- Prone to IP conflicts if not managed
- Not ideal for mobile devices

12. What is DHCP and how does it work?

- Answer: DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses, subnet masks, default gateways, and DNS settings to devices on a network.

☒ **Advanced/Scenario-Based Interview Questions**

13. Can you assign a static IP within a DHCP range?

- Answer: Yes, but it's not recommended unless reserved via DHCP to avoid conflicts.

14. What happens if two devices have the same IP address?

- Answer: An IP conflict occurs, which may cause connectivity issues or network errors.

15. How does NAT work with private IP addresses?

- Answer: NAT (Network Address Translation) allows multiple private IPs to share one public IP for internet access by translating the internal private IP to a public one.

16. Can a device have both public and private IP addresses?

- Answer: Yes. For example, in cloud environments, an instance may have both for internal and external communication.

☒ Quick Reference Summary

Term	Description
Public IP	Routable on internet, unique globally
Private IP	For internal networks, non-routable
Static IP	Fixed and manually assigned
Dynamic IP	Assigned automatically, may change
DHCP	Protocol to assign IPs dynamically
NAT	Translates private IPs to public ones

☒ Amazon VPC (Virtual Private Cloud) Interview Questions

1. What is a VPC in AWS?

- Answer: VPC is a logically isolated network in AWS where you can launch AWS resources. It gives control over IP ranges, subnets, route tables, internet gateways, NAT, and firewalls.

2. What are the main components of a VPC?

- Answer:

- Subnets (Public/Private)
- Route Tables
- Internet Gateway (IGW)
- NAT Gateway/Instance
- Network ACLs
- Security Groups
- DHCP Option Sets

3. What is the default VPC?

- Answer: A pre-configured VPC AWS creates in each region. It includes:
 - One subnet per AZ
 - Internet gateway attached
 - Public IP auto-assigned
 - Route table for internet access

4. How is a custom VPC different from a default VPC?

- Answer: A custom VPC is user-defined with manually created subnets, route tables, etc., whereas a default VPC is automatically set up.

5. **Can you have multiple subnets in the same AZ in a VPC?**

- Answer: Yes, but they must have non-overlapping CIDR blocks.

VPC Peering Interview Questions

6. **What is VPC peering in AWS?**

- Answer: A networking connection between two VPCs that enables routing traffic using private IPs as if they were on the same network.

7. **Can VPC peering work across regions or accounts?**

- Answer: Yes. VPC peering supports inter-region and cross-account connections.

8. **What are limitations of VPC peering?**

- Answer:
 - No transitive peering
 - No overlapping CIDR blocks
 - No security group referencing across VPCs (unless using PrivateLink or Transit Gateway)

9. **What must be updated after a peering connection is established?**

- Answer:
 - Route tables in both VPCs
 - (Optionally) Network ACLs and Security Groups for traffic flow

10. What is the difference between VPC peering and AWS Transit Gateway?

- Answer:
 - *VPC Peering*: Point-to-point
 - *Transit Gateway*: Central hub for many VPCs, supports transitive routing
-



AWS CLI Interview Questions

11. What is the AWS CLI?

- Answer: The AWS Command Line Interface (CLI) allows interaction with AWS services using commands in a terminal or script.

12. How do you configure the AWS CLI?

Answer:

```
#aws configure
```

- Then input:
 - Access Key
 - Secret Key
 - Default region
 - Output format (e.g., **json**, **table**, **text**)

13. How do you create a VPC using AWS CLI?

Answer:

```
#aws ec2 create-vpc --cidr-block 10.0.0.0/16
```

14. How do you view all running EC2 instances using CLI?

Answer:

```
#aws ec2 describe-instances --filters Name=instance-state-name,Values=running
```

15. How do you create a VPC peering connection using CLI?

Answer:

```
#aws ec2 create-vpc-peering-connection \  
--vpc-id vpc-12345678 \  
--peer-vpc-id vpc-87654321 \  
--peer-owner-id 123456789012
```

16. How do you add a route in a route table using CLI?

Answer:

```
#aws ec2 create-route \  
--route-table-id rtb-12345678 \  
--destination-cidr-block 10.1.0.0/16 \  
--vpc-peering-connection-id pcx-12345678
```

17. How do you list all VPCs using CLI?

Answer:

```
#aws ec2 describe-vpcs
```

⚡ Scenario-Based / Practical Questions

18. You launched an EC2 instance but can't connect. How would you troubleshoot using CLI?

- Answer:
 - Check security group rules: `aws ec2 describe-security-groups`
 - Verify route tables and subnet settings
 - Check VPC and subnet CIDR overlaps

- Validate instance state and public IP

19. How do you automate VPC creation and resource provisioning?

- Answer:
 - Use AWS CLI scripts
 - Use CloudFormation templates
 - Or use Terraform (third-party)

20. Can you use CLI to connect to EC2 instance?

- Answer: No. You use CLI to manage the instance, but SSH (or PuTTY) is used to connect.

☒ Basic IAM Interview Questions

1. What is IAM in AWS?

- Answer: IAM (Identity and Access Management) is a service that helps securely control access to AWS services and resources for users, groups, and roles.

2. What are IAM users, groups, and roles?

- Answer:
 - User: An individual identity with long-term credentials (username/password or access keys).

- Group: A collection of IAM users; used to assign permissions collectively.
- Role: A temporary identity assumed by users, services, or other AWS accounts; it has a defined set of permissions.

3. What are IAM policies?

- Answer: JSON documents that define permissions. They are attached to users, groups, or roles to control access.

4. What are the types of IAM policies?

- Answer:
 - Managed policies: AWS-managed or customer-managed.
 - Inline policies: Embedded directly into a user, group, or role.

Intermediate IAM Interview Questions

5. What is the difference between IAM role and IAM user?

- Answer: IAM users have long-term credentials; roles are assumed temporarily and do not have permanent credentials.

6. What is a trust policy?

- Answer: Defines who can assume a role (e.g., another user, account, or service).

7. How can you enforce MFA (Multi-Factor Authentication) in IAM?

- Answer: Attach a policy condition like `"Condition": {"Bool": {"aws:MultiFactorAuthPresent": "true"}}` to require MFA.

8. How do IAM permissions work?

- Answer: IAM evaluates all allow and deny statements across attached policies. Explicit deny overrides allow.

9. What is the IAM policy simulator?

- Answer: A tool to test and troubleshoot IAM policies and verify whether access is allowed or denied.

Advanced IAM Interview Questions

10. How would you allow cross-account access using IAM?

- Answer: Create a role in Account A with a trust policy for Account B, then allow users in Account B to assume that role.

11. What is the difference between resource-based and identity-based policies?

- Answer:
 - Identity-based: Attached to users, groups, or roles.
 - Resource-based: Attached to AWS resources (e.g., S3 bucket policy, Lambda resource policy).

12. What are IAM conditions in policies?

- Answer: Conditions refine permissions using keys (e.g., IP range, time of day, MFA present).

Example:

```
"Condition": {  
  "IpAddress": {  
    "aws:SourceIp": "203.0.113.0/24"  
  }  
}
```

13. How would you restrict access to an S3 bucket to a specific VPC?

- Answer: Use a bucket policy with the condition:

```
"Condition": {  
  "StringEquals": {  
    "aws:SourceVpc": "vpc-abc123"  
  }  
}
```

14. What are AWS Organizations and SCPs (Service Control Policies)?

- Answer:

- AWS Organizations allows centralized management of multiple accounts.
 - SCPs restrict maximum available permissions across accounts, overriding IAM policies.
-

Scenario-Based IAM Questions

15. You granted an IAM user full S3 access, but access is denied. What could be the issue?

- Answer:

- A bucket policy or SCP explicitly denies access.
- Missing region or credentials misconfigured.
- MFA required but not provided.

16. How can you give an EC2 instance permission to access an S3 bucket?

- Answer:

- Attach an IAM role to the EC2 instance.
- The role must have a policy granting access to the S3 bucket.

17. How do you rotate IAM access keys securely?

- Answer:
 - Create a second access key.
 - Test it.
 - Deactivate and delete the old key.

18. How can you audit IAM activity?

- Answer:
 - Use AWS CloudTrail to track API calls.
 - Use IAM Access Analyzer to identify external access.

19. How do you limit a user to launch only t2.micro EC2 instances?

- Answer: Use a condition in the policy like:

```
"Condition": {  
  "StringEquals": {  
    "ec2:InstanceType": "t2.micro"  
  }  
}
```

Quick IAM Facts Cheat Sheet

Feature	Description
IAM Role	Temporary credentials, assumed by users/services
IAM User	Permanent credentials (access key/password)
Managed Policy	AWS or customer managed, reusable policies
Inline Policy	Embedded directly in a user/group/role
MFA	Adds second layer of authentication
SCP	Restricts permissions at the Org/account level
IAM Analyzer	Finds unintended access to resources

☒ Basic Cron Job Interview Questions

1. What is a cron job?

- Answer: A cron job is a scheduled task in Unix/Linux systems that runs automatically at specified times or intervals.

2. Where are cron jobs defined?

- Answer: In the crontab file, accessed using `crontab -e` for user-specific jobs, or in system-wide cron directories like `/etc/crontab`, `/etc/cron.daily`.

3. What is the syntax of a cron job?

- Answer:

`* * * * * /path/to/command`

`| | | | |`

`| | | | +----- Day of the week (0-7) (Sun=0 or 7)`

`| | | +----- Month (1-12)`

`| | +----- Day of month (1-31)`

`| +----- Hour (0-23)`

`+----- Minute (0-59)`

4. How do you list current user's cron jobs?

- Answer:

`crontab -l`

5. How do you edit a user's crontab?

- Answer:

`crontab -e`

◇ Intermediate Cron Job Interview Questions

6. What is the difference between user crontab and system crontab?

- Answer:

- *User crontab*: Managed with `crontab -e`, does not include the user field.
- *System crontab (/etc/crontab)*: Includes an extra field to specify the user under which the job runs.

7. What does the following cron job do?

```
0 2 * * * /home/user/backup.sh
```

- Answer: Runs `backup.sh` every day at 2:00 AM.

8. How can you ensure output or errors from a cron job are logged?

- Answer:
Redirect output like this:

```
* * * * * /path/to/script.sh >> /var/log/script.log 2>&1
```

9. How do you schedule a job to run every 15 minutes?

- Answer:

```
*/15 * * * * /path/to/script.sh
```

10. How do you run a job every Monday at 6 PM?

- Answer:

```
0 18 * * 1 /path/to/script.sh
```

△ Advanced Cron Job Interview Questions

11. How can you prevent overlapping cron job executions?

- Answer:

Use a lock file or wrapper scripts like:

```
/usr/bin/flock -n /tmp/lock.file /path/to/script.sh
```

-
- Use cronlock or tools like [jobber](#).

12. How do you temporarily disable a cron job?

- Answer:

- Comment out the line with `#` in the crontab.
- Use `crontab -r` to remove all jobs (not recommended without backup).

13. Why might a cron job run manually but not via cron?

- Answer:

- Environment variables are different in cron.
- Permissions issues.

- Paths to executables not set (use full paths).
- Scripts lack executable permission.

14. How do you set environment variables in a crontab?

- Answer: Add them at the top of the crontab file:

`SHELL=/bin/bash`

`PATH=/usr/bin:/bin:/usr/sbin:/sbin`

15. How would you schedule a job to run at reboot?

- Answer:

`@reboot /path/to/script.sh`

⚡ Scenario-Based Cron Job Questions

16. A cron job logs "command not found" — how do you debug it?

- Answer:
 - Use full paths to commands.
 - Check `PATH` variable in cron.
 - Redirect output to log files.

17. How do you monitor cron job failures?

- Answer:
 - Redirect output to a log.
 - Configure cron to send email (`MAILTO=user@example.com`).
 - Use monitoring tools like Nagios, CloudWatch (in AWS), or custom log monitoring.

18. How would you run a job every weekday (Mon–Fri) at 8 AM?

- Answer:

```
0 8 * * 1-5 /path/to/script.sh
```

19. How would you run a script on the last day of every month?

- Answer: Cron can't do this natively. Instead:

- Use a script to check if today is the last day:

```
[ "$(date +%d -d tomorrow)" == "01" ] && /path/to/script.sh
```

- Schedule it to run daily.

20. How do you write a cron job to run every 10 minutes from 9 AM to 5 PM?

- Answer:

```
*/10 9-17 * * * /path/to/script.sh
```

Bonus: Crontab Special Strings

String	Meaning
@reboot	Run at system startup
@hourly	Run once per hour
@daily	Run once a day
@weekly	Run once a week
@monthly	Run once a month
@yearly	Run once a year

Basic AWS Lambda Interview Questions

1. What is AWS Lambda?

- Answer: AWS Lambda is a serverless compute service that runs code in response to events without provisioning or managing servers. You pay only for the compute time consumed.

2. What languages does AWS Lambda support?

- Answer:

- Natively: Python, Node.js, Java, Go, Ruby, .NET (C#), PowerShell
- Custom runtimes can be used for other languages via AWS Lambda Runtime API.

3. How is AWS Lambda billed?

- Answer: Based on:
 - Number of requests
 - Duration (in milliseconds), rounded up to the nearest ms
 - Memory allocated (128MB to 10GB)
 - Optional: Provisioned concurrency

4. What are the main use cases for Lambda?

- Answer:
 - Real-time file processing (e.g., S3 uploads)
 - API backends (via API Gateway)
 - Event-driven tasks (e.g., DynamoDB streams, SNS)
 - Scheduled jobs (cron with EventBridge)

- Chatbots, microservices, data transformation

5. What triggers can invoke a Lambda function?

- Answer:
 - S3, DynamoDB, SNS, SQS, CloudWatch Events, API Gateway, EventBridge, Step Functions, etc.
-

Intermediate AWS Lambda Interview Questions

6. What is the maximum timeout for a Lambda function?

- Answer: 15 minutes (900 seconds).

7. How do you pass environment variables to a Lambda function?

- Answer: Use the Lambda console, AWS CLI, or CloudFormation to define environment variables.

8. What is cold start in AWS Lambda?

- Answer: A cold start is the delay that occurs when AWS provisions a new container to run your Lambda function, typically during the first invocation or after a period of inactivity.

9. What is the Lambda execution role?

- Answer: An IAM role assumed by Lambda at runtime to access AWS services (e.g., S3, DynamoDB).

10. How can you secure sensitive data in a Lambda function?

- Answer:
 - Store secrets in AWS Secrets Manager or SSM Parameter Store
 - Encrypt environment variables using KMS
 - Use least privilege IAM roles
-

Advanced AWS Lambda Interview Questions

11. What is provisioned concurrency in Lambda?

- Answer: Keeps a set number of Lambda instances initialized and ready to handle traffic instantly, reducing cold starts.

12. Can a Lambda function call another Lambda?

- Answer: Yes, directly via AWS SDK or indirectly via services like Step Functions, EventBridge, or SNS.

13. How do you monitor Lambda performance and logs?

- Answer:
 - Amazon CloudWatch Logs: Stores stdout/stderr logs
 - CloudWatch Metrics: Invocation count, duration, error count

- X-Ray: Distributed tracing

14. How do you handle retries and errors in Lambda?

- Answer:
 - Automatic retries for async invocations (2 retries by default)
 - Use Dead Letter Queues (DLQ) for failure handling
 - Implement try-catch logic for sync invocations
 - Use Lambda Destinations for async success/failure handling

15. What's the difference between synchronous and asynchronous Lambda invocation?

- Answer:
 - Synchronous: Caller waits for result (e.g., API Gateway)
 - Asynchronous: Event is queued and processed later (e.g., S3, SNS)

Scenario-Based Lambda Questions

16. Your Lambda function is timing out — how do you troubleshoot?

- Answer:

- Check CloudWatch Logs for execution time
- Increase the timeout setting if necessary
- Optimize code, check downstream services
- Use X-Ray for tracing bottlenecks

17. How would you handle batch processing with Lambda?

- Answer:
 - Use SQS, Kinesis, or DynamoDB Streams to trigger Lambda
 - Set batch size appropriately
 - Use partial batch failure handling (supported in SQS, Kinesis)

18. How can you version or deploy Lambda functions safely?

- Answer:
 - Use versions and aliases
 - Deploy with CodeDeploy for traffic shifting
 - Use CI/CD pipelines (CodePipeline, GitHub Actions, etc.)

19. How do you share code across multiple Lambda functions?

- Answer:
 - Use Lambda Layers
 - Package shared libraries as dependencies

20. How do you limit concurrent executions in Lambda?

- Answer:
 - Set Reserved Concurrency for the function
 - Use Account-level concurrency limits to control cost and resource use

Bonus: Quick Facts

Feature	Limit / Info
Timeout max	15 minutes (900 sec)
Memory range	128 MB – 10 GB
Max package size	50 MB (zip), 250 MB (unpacked)
Execution environment	Ephemeral / stateless

Temporary storage	<code>/tmp</code> directory (512 MB default, 10 GB with configuration)
Deployment methods	Console, CLI, SDK, SAM, CDK, Terraform
Runtime updates	Managed by AWS

☒ Basic RDS Interview Questions

1. What is Amazon RDS?

- Answer: Managed relational database service that simplifies setup, operation, and scaling of databases in the cloud.

2. Which database engines does RDS support?

- Answer:
 - Amazon Aurora (MySQL and PostgreSQL compatible)
 - MySQL
 - PostgreSQL
 - MariaDB
 - Oracle
 - Microsoft SQL Server

3. What are the benefits of using RDS?

- Answer:
 - Automated backups and snapshots
 - Automatic software patching
 - Scalability
 - High availability with Multi-AZ deployments
 - Security integration with IAM and KMS
 - Monitoring via CloudWatch

4. What is a Multi-AZ deployment in RDS?

- Answer: Synchronous replication to a standby instance in another Availability Zone for high availability and failover.

5. What are read replicas in RDS?

- Answer: Asynchronous copies of the primary DB instance used to offload read traffic and improve performance.

Intermediate RDS Interview Questions

6. What is the difference between Multi-AZ and Read Replica?

- Answer:
 - *Multi-AZ*: For high availability and failover, synchronous replication.
 - *Read Replica*: For scaling read workload, asynchronous replication.

7. How does RDS handle backups?

- Answer: Automated backups are stored in S3, can be retained up to 35 days. Manual snapshots persist until deleted.

8. Can you modify the instance class or storage type of an RDS database?

- Answer: Yes, you can scale up/down the instance type and change storage (e.g., from magnetic to SSD).

9. How do you secure data in RDS?

- Answer:
 - Encrypt data at rest with AWS KMS
 - Use SSL/TLS for data in transit
 - Configure security groups and subnet groups
 - IAM database authentication (for supported engines)

10. What is a DB parameter group?

- Answer: A configuration container for database engine settings that can be applied to RDS instances.

⚙️ Advanced RDS Interview Questions

11. What are some limitations of RDS?

- Answer:
 - Limited superuser access (managed service)
 - Limited control over OS-level settings
 - Some engines may not support all features available in self-managed versions.

12. How can you monitor RDS performance?

- Answer:
 - Amazon CloudWatch metrics (CPU, memory, disk I/O)
 - Enhanced Monitoring for OS-level metrics
 - Performance Insights for query analysis

13. What is Aurora and how is it different from standard MySQL or PostgreSQL on RDS?

- Answer:

- Aurora is a proprietary AWS engine optimized for cloud with distributed storage, better scalability, and faster recovery.

14. How does failover work in Multi-AZ RDS?

- Answer: AWS automatically switches to standby instance with minimal downtime during failure or maintenance.

15. How would you migrate a database to RDS?

- Answer:
 - Use AWS Database Migration Service (DMS)
 - Use native backup/restore or dump/import tools
 - Replication-based migration (read replicas)
-

Scenario-Based RDS Questions

16. Your RDS instance is running slow. How do you troubleshoot?

- Answer:
 - Check CloudWatch metrics for CPU, disk, IOPS
 - Review slow query logs or enable Performance Insights
 - Check instance size and storage type

- Check for locks or long-running queries

17. How do you perform a backup and restore in RDS?

- Answer:

- Use automated backups or create manual snapshots
- Restore by launching a new instance from a snapshot

18. How do you handle database failover with minimal downtime?

- Answer: Use Multi-AZ deployments for automatic failover.

19. Can you connect to an RDS instance from outside AWS?

- Answer: Yes, if the instance is in a public subnet with proper security group and network ACLs.

20. How do you ensure data security and compliance with RDS?

- Answer:

- Use encryption at rest and in transit
- Manage access with IAM and security groups
- Enable auditing/logging
-

Apache HTTP Server Interview Questions

1. What is Apache HTTP Server?

- An open-source web server software that serves web content using the HTTP protocol.

2. How do you start, stop, and restart Apache server?

Commands vary by OS; for example, on Linux:

```
#sudo systemctl start apache2
```

```
#sudo systemctl stop apache2
```

```
#sudo systemctl restart apache2
```

-

3. What are Apache modules? Name some common modules.

- Modules extend Apache functionality (e.g., mod_rewrite, mod_ssl, mod_proxy).

4. How do you enable or disable Apache modules?

- Using **a2enmod** and **a2dismod** commands on Debian/Ubuntu.

5. What is .htaccess?

- A per-directory configuration file used to override Apache server settings.

6. How do you configure virtual hosts in Apache?

- Using **<VirtualHost>** blocks to host multiple sites on a single server/IP.

7. How does Apache handle multiple requests?

- Apache supports process-based and thread-based models (prefork, worker, event MPM).

8. How do you set up SSL/TLS in Apache?

- Enable `mod_ssl`, configure `SSLCertificateFile`, `SSLCertificateKeyFile` in config.

9. What is `mod_rewrite` used for?

- URL rewriting and redirection.

10. How can you improve Apache server performance?

- Enable caching, tune MPM settings, use KeepAlive, offload static content.

Nginx Interview Questions

1. What is Nginx?

- A high-performance HTTP and reverse proxy server known for handling large numbers of concurrent connections efficiently.

2. How is Nginx different from Apache?

- Event-driven (asynchronous) vs process/thread-based model in Apache, resulting in better performance under high load.

3. What are the main use cases for Nginx?

- Web serving, reverse proxy, load balancing, caching, API gateway.

4. How do you configure server blocks (virtual hosts) in Nginx?

- Using `server {}` blocks inside config files.

5. What is the purpose of the `worker_processes` and `worker_connections` directives?

- They control concurrency and how many requests can be handled simultaneously.

6. How do you enable SSL in Nginx?

- Configure `ssl_certificate` and `ssl_certificate_key` directives in the server block.

7. What is reverse proxy and how does Nginx implement it?

- Nginx forwards client requests to backend servers and serves the response.

8. How do you handle load balancing in Nginx?

- Use the `upstream {}` directive to define backend servers and balance methods like round-robin.

9. What caching mechanisms does Nginx support?

- Proxy caching, FastCGI caching, microcaching.

10. How can you improve Nginx performance?

- Optimize worker processes, enable caching, use gzip compression, tune buffers.

Apache vs Nginx: Common Interview Comparisons

Feature	Apache	Nginx
Architecture	Process/thread-based	Event-driven asynchronous
Handling concurrent connections	Uses more memory under load	Highly efficient, low memory
Static content serving	Good	Excellent
Dynamic content support	Supports via modules (e.g., mod_php)	Relies on backend (FastCGI)
Configuration	<code>.htaccess</code> supports per-directory overrides	No <code>.htaccess</code> support
Load balancing	Basic, via modules	Built-in, advanced
Reverse proxy	Supported via modules	Core feature